

The Harness Is the Edge: Leak-Free Evaluation of a Disciplined Language-Model Agent on Weather Prediction Markets

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Abstract

In the one published benchmark of frontier language models trading real prediction markets with real capital, all six models lost money. We argue the failure is a *harness* problem more than a forecasting one, and test that claim where it is measurable: daily station-temperature markets on Kalshi and Polymarket, which settle every day against the U.S. National Weather Service report for a named station. **OpenThomas**, an open-source agent, prices each strike from a seven-model NWP consensus with a per-station bias learned from leak-free hindcasts, lets a language model make a *bounded* adjustment, blends with the market prior, and sizes with fractional Kelly under risk limits the model cannot override. Our main contribution is methodological: a leak-free replay on real order books with explicit windows, frozen and hashed datasets, local-time decision snapshots, official NBM/MOS baselines, day-block bootstrap error bars, and a *pre-registered* ablation whose control decides whether the edge is selectivity or a better probability. A preliminary 21-day replay had found a market-prior blend positive (+\$0.028/contract). The pre-registered replay over 64 settlement days (3,349 markets) does not: every setting of the baseline-only rule loses (−\$0.034/contract, 95% CI [−0.069, +0.002]), the blend and a raised edge bar are the same lever, and the consensus with learned bias ties the NWS National Blend of Models (Brier 0.163 vs. 0.163). The bounded LLM adjustment improves Brier slightly and the clamp sits at the best of a sweep in which the unclamped model does worse. What survives is the harness: the drawdown kill-switch is the one rail that preserved capital under a losing rule, and the self-improvement gate refused thin evidence and rolled one promotion back. We report the result that contradicts our own thesis, and what it changes. All data, frozen replay rows, and results are published daily.

Keywords

prediction markets, large language model agents, autonomous trading, forecast calibration, numerical weather prediction, backtesting, risk management

Status of this draft. This paper is written build-in-public. The experiments of Section 7 are pre-registered with their decision rules, written before the data; every number is filled from a daily pipeline over a public, growing dataset and is regenerated as the window rolls forward. The first run contradicted the thesis, and the thesis was rewritten, not the data. Any cell that reads **TBD**: is awaiting the next run.

1 Introduction

The obvious way to build a trading agent on top of a large language model (LLM) is to ask the model what will happen and bet accordingly. The empirical record so far says this does not work. *Prediction Arena* [1] gave six frontier models \$10,000 each and let them trade real prediction markets autonomously for 57 days. All six lost money, between −16% and −30.8%. Research *volume* had

no correlation with returns; token spend had none either. What separated the least-bad runs from the worst was not a better world model but better *behavior*: being right early, sizing up when confident and down when uncertain, holding winners to settlement, and, above all, not trading when there was no edge. The single profitable run on record (+10.9%) was selective (112 trades), asymmetric (average win \$63.89 vs. average loss \$3.23), and low-drawdown (4.1%). One benchmark of six models over 57 days is thin evidence for a universal law; it is, however, exactly the profile a practitioner would predict, and it motivates the hypothesis this paper sets out to test rather than assert:

The LLM is not the edge. The harness is the edge.

If the edge lives in the harness, progress needs a domain where harness quality is *measurable*: densely, objectively, and without leakage. Most prediction markets fail this test: they settle rarely, their resolution is ambiguous, and any backtest risks letting the replayed market “know” something the replayed model did not. **Weather markets are the exception.** A daily station-temperature market (“Will the high in Central Park exceed 83°F today?”) resolves against the NWS climatological report for one named station, every day, with no interpretive latitude. Dozens of ground-truth data points arrive per day.

We present **OpenThomas**,¹ an open-source (≈8,000 lines of Python, 213 tests) agent that operationalizes the harness thesis on weather markets, and, more importantly, a way of evaluating it that we believe is the honest one. The paper is written *build-in-public*: the experiments in Section 7 are pre-registered with their decision rules, the numbers are filled from a daily pipeline over a growing public dataset, and the conclusions follow the numbers; where they contradicted our thesis, we changed the thesis.

Design commitments:

- **Selectivity over activity.** “No trade” is the default. The agent trades only when $|\hat{p} - \text{price}|$ clears an explicit expected-value threshold after fees and spread.
- **A statistical baseline the model may adjust but never replace.** Every strike is priced from a seven-model NWP consensus, corrected by per-(station, lead) bias and spread learned from leak-free hindcasts, and hard-truncated by what today’s thermometer has already locked in. The LLM is clamped to ± 0.15 around this baseline.
- **Hard risk constraints outside the model.** Fractional Kelly sizing, per-market / per-event / per-category caps, a fee-aware EV threshold, a longshot-zone filter, and a drawdown kill-switch are deterministic code.
- **Calibration as a first-class metric,** tracked per (station, lead) as Brier skill against the market.
- **Self-improvement as a gated loop** with the evaluator structurally off the agent’s edit surface (Section 5).

¹Open-source; the repository, live project and dataset URLs are withheld to preserve double-blind anonymity and will be restored in the camera-ready version.

We contribute: (1) the harness-is-the-edge thesis as a concrete system and a *pre-registered, falsifiable* test of it (Section 7, E1); (2) a leak-free hindcast/replay methodology for market agents, with explicit windows, frozen and hashed datasets, local-time snapshots, official-guidance baselines (NBM, GFS MOS) and day-block bootstrap error bars (Section 6); and (3) a self-improvement architecture whose safety comes from a kernel/agent plane split that keeps the judge off the edit surface, with its empirical record (promotions, rollbacks, vetoes) reported as it accrues (Section 5).

2 Background and related work

LLMs as forecasters and traders. A growing literature evaluates LLMs as probabilistic forecasters. Retrieval-augmented systems approach, but do not exceed, aggregate human forecasters [3]; LLM ensembles match crowd accuracy on some question sets [4]; and the consistent practical finding is that a *pure* LLM forecast underperforms the market while a *blend* of model and market price can beat either [2]. Prediction Arena [1] extends this to live autonomous trading and finds uniform losses, isolating behavioral rather than epistemic failure. We take the blend result and the behavioral diagnosis as design inputs, and we test, rather than assume, which of the two carries the edge (E1).

Statistical post-processing of weather forecasts. Learning a per-station correction to model output is Model Output Statistics [5], in operational use for fifty years; the NWS’s National Blend of Models (NBM) [6] is a bias-corrected multi-model blend published for exactly the stations these markets settle on, and GFS MOS bulletins are archived for every cycle. Our baseline is a small MOS: a shrunken mean error per (station, kind, lead) on a seven-model consensus. Whether it adds anything over what the market already reads is an empirical question we answer by pricing the same markets from NBM and GFS MOS (E5) rather than assuming a “local edge.” Machine-learned global models now match or exceed physics-based systems: GraphCast [7] gives deterministic 10-day forecasts, GenCast [8] calibrated ensembles via diffusion. OpenThomas runs both on commodity accelerators; GraphCast is a voter in the consensus and GenCast’s ensemble spread is a flow-dependent multiplier on the baseline’s σ .

Sizing, calibration, microstructure. Fractional Kelly sizing [9] and Platt scaling [10] are standard; we use them as such. The favorite-longshot bias [11] is encoded as a hard price-zone filter. Cross-venue listing of the same event (Polymarket vs. Kalshi) creates episodic arbitrage, which the scanner flags deterministically.

Self-improving agents. A recent framing of near-term recursive self-improvement (RSI) locates the leverage in the *harness* and names evaluator strength as the first bottleneck [12]. The Darwin Gödel Machine [13] keeps an archive of past variants as mutation parents. OpenThomas adopts both and adds a security model: because weather markets supply an unusually strong evaluator, the risk is a *gameable* judge, so the judge is made structurally unwritable by the optimizer.

3 Why weather markets are a strong-evaluator domain

Dense, objective settlement. Each market resolves against the NWS Climatological Report (product CLI) for one named station, once per day: a single integer, the day’s observed maximum or minimum. A generalist agent might see a few dozen settlements per *year*; a weather desk on ten U.S. cities (highs and lows) sees dozens per *day*.

A learnable, local correction. Markets settle on a *specific* station (NYC = Central Park; Chicago = Midway), and global models have systematic, station-specific errors there. Throughout, bias is defined as $b = \text{settled} - \text{consensus}$, so a positive b means the models run *cold*. Our first 90-day hindcast found Miami $b \approx +1.8$ to $+2.1^\circ\text{F}$ (σ 1.7), Philadelphia $+2.3$, and LAX ≈ -1 (models warm). Whether these persist (E9) and whether they beat NBM’s own correction (E5) are measured, not assumed.

Leak-free reproducibility. Historical “as-of” forecasts (what a model published at a past issue time) and historical order books are both available, so a backtest can present the replayed agent with exactly what it would have had, and no more. That is what lets us evaluate the *decision rule*, and later self-improvement, without the replay silently cheating. These are the properties the RSI literature asks for and rarely gets [12].

4 System architecture

OpenThomas is a stack of layers with a strict rule about who is in control at each one (Figure 1, Table 1). Everything that touches money is deterministic code; the language model is confined to a bounded adjustment in the middle of the trade path.

Market connectors and paper simulation. A unified Market/Order/Position model spans Polymarket (Gamma API for discovery, CLOB for books/orders) and Kalshi (REST + WebSocket, RSA-signed, CFTC-regulated). A *paper* connector wraps either venue’s real market data but simulates fills locally at bid/ask (never mid), with mark-to-market at bid (liquidation value). Paper mode is the default and needs no exchange keys.

The weather baseline. For a market on station s , valid day d , kind $k \in \{\text{high, low}\}$, the baseline forms a consensus temperature $\bar{T}$ across seven operational global models (GFS, ECMWF-IFS, ICON, GEM, UKMO, Météo-France, JMA) plus the system’s own GraphCast point forecast, with cross-model spread σ_{model} , and applies two corrections learned strictly out-of-window:

$$\hat{T} = \bar{T} + b_{s,k,\ell}, \quad \sigma = \max(\sigma_{s,k,\ell}, 0.8 \sigma_{\text{model}}) \cdot g_{s,d},$$

where ℓ is the forecast lead, $b_{s,k,\ell}$ the learned bias, $\sigma_{s,k,\ell}$ the residual spread (both shrunk toward a climatological prior with pseudo-count 10), and $g_{s,d} \in [0.7, 1.6]$ the GenCast flow-dependent multiplier (the day’s ensemble spread relative to a typical day at that lead; GenCast’s 12-hour steps undersample the diurnal peak, so it contributes uncertainty, not a point estimate). The per-strike probability is the mass on the correct side of the half-integer strike under a fat-tailed mixture ($0.8 \mathcal{N}(\hat{T}, \sigma) + 0.2 \mathcal{N}(\hat{T}, 2\sigma)$), with one more hard step: *truncation*. If today’s thermometer has already made one side impossible, the probability is clamped accordingly. This is

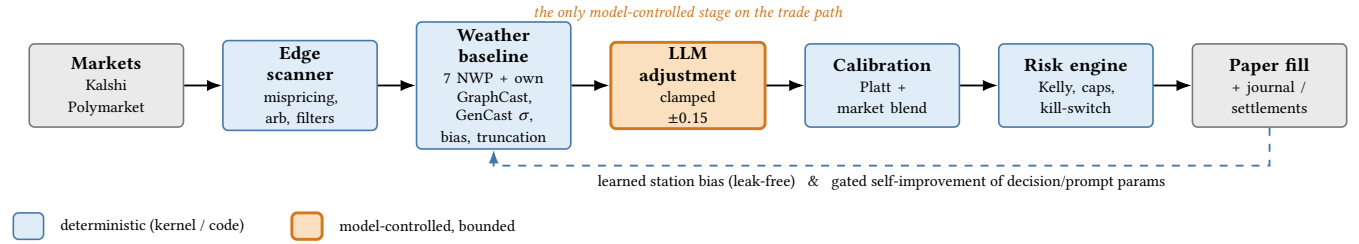


Figure 1: The OpenThomas pipeline. A trade flows left to right; the one model-controlled stage on the trade path is clamped to the statistical baseline ± 0.15 and bracketed by deterministic layers. (The post-settlement reflection that distills playbook rules is also model-written, but off the trade path and human-auditable.) Settled outcomes feed back to teach the baseline each station’s bias (strictly leak-free) and to gate self-improvement of decision and prompt parameters. Sizing and exposure never leave deterministic code.

Layer	What it does	Who’s in control
Edge scanner	Filters markets to plausible mispricings; flags cross-venue arbitrage and longshot / resolution-trap zones	deterministic code
Weather baseline	Seven-model NWP consensus + own GraphCast $\rightarrow$ per-strike probability at the settlement station, with per-(station, kind, lead) bias/ σ from leak-free hindcasts, GenCast spread as a σ multiplier, and hard truncation by today’s observed extreme	deterministic code
Forecast engine	LLM ensemble (median of N independent estimates), grounded in resolution rules and base rates, <i>clamped to baseline</i> ± 0.15	your choice of model
Calibration	Platt-scales forecasts against your own settled history; blends with the market prior	deterministic code
Risk engine	Fractional Kelly, per-market/event/category caps, fee-aware EV threshold, longshot filter, drawdown kill-switch	deterministic code
Memory	Every forecast and fill journaled to SQLite; post-settlement reflection distills bounded playbook rules	agent, human-auditable

Table 1: The OpenThomas stack. The model *proposes*; the risk engine *disposes*. One layer on the trade path is model-controlled, bounded above and below by deterministic layers.

where a naive backtest leaks, and where Section 6 spends most of its care.

Forecast engine: bounded LLM adjustment. The model receives the baseline probability, resolution rules, base rates, and a news brief, and returns an adjusted probability, then *clamped to baseline* ± 0.15 . The statistics do the heavy lifting; the model is there for what the statistics cannot see (a blown front, a stale consensus, a discontinuity in the question text). Because of the clamp, mid-size local reasoning models suffice, which keeps the agent runnable on self-hosted open-weights models. Whether the adjustment adds anything, and at which clamp, is E4.

Calibration and market-prior blend. Once ≥ 30 markets settle in a category, OpenThomas fits Platt scaling, then blends the calibrated forecast with the market price: $p = w \cdot \text{mid} + (1 - w) \cdot \hat{p}$, $w = 0.5$ in production. The blend weight and the minimum edge are entry-selectivity parameters and are candidates for self-improvement under kernel bounds (Section 5); sizing is not.

Risk engine (deterministic). The engine takes bankroll, risk profile, the blended probability, and market price/liquidity, and returns an approved size or a rejection naming the binding constraint: fractional Kelly (0.15 $\times$ to 0.33 $\times$ by profile), per-market (5%), per-event (10%) and per-category (25%) caps at cost basis, 50% total open

risk, a fee-aware EV threshold, a $[0.05, 0.95]$ price zone, a solvency check, and a drawdown kill-switch (15%) that halts trading until a human resumes. None of these are reachable by the model or by the self-improvement loop; a kernel policy test fails any change that tries to move a safety rail into the optimizer’s reach. What each rail costs or saves on the replayed trades is E7.

5 Self-improvement with the judge off the edit surface

OpenThomas updates OpenThomas. The operator is moved *up* the stack, setting bounds and reviewing lineage, and is forbidden from hand-tuning trading parameters. The design problem is that self-improvement is safe only if improvement cannot redefine “improvement”: an optimizer that can edit its own judge will find it easier to lower the bar than to clear it.

Kernel/agent planes. The codebase is split into two planes, and the split is the security model (Table 2). The **kernel** plane (promotion gate, parameter bounds, the risk engine and all safety rails, the truth pipeline, the journal schema) is written only by the operator; the evolution loop has no write path to it. The **agent** plane (decision-rule parameters, the forecast prompt template; on the roadmap, workflow structure and strategy code) is written by the evolution loop, only *through* the kernel gate. Entry-selectivity knobs

Plane	Contents	Written by
kernel	bounds, promotion gate, risk engine & safety rails, truth pipeline, journal schema	operator only
agent	decision-rule params, forecast prompt template; later: workflow, strategy code, the evolver	evolution loop, <i>via the gate</i>

Table 2: The plane split is the security model: the judge is not on the optimizer’s edit surface.

(minimum edge, blend weight) are genome parameters under kernel bounds; sizing, exposure caps and the kill-switch can never enter the parameter space.

Loops and the gate. A fast *trading* loop (minutes) runs the champion parameters. A *reflection* loop (on settlements) distills bounded playbook rules. An *evolution* loop mines failure evidence from the journal, proposes mutations (LLM-directed, plus Gaussian mutations from an archive of past variants [13]) and submits them to the gate, which scores a candidate *strategy callable* on leak-free replay with four anti-gaming properties: *held-in/held-out* splits (winners are picked on older days; the freshest seven days only veto, and the window rolls forward daily); *totals-not-ratios* with a minimum trade count (a rule that trades twice cannot win on ratio; one that trades nothing cannot win on drawdown); *out-of-sample regret roll-back* (each cycle re-scores the active generation against its parent on the fresh window and reverts if the parent now wins); and a *Brier veto* for the forecast operator (a prompt that wins PnL while worsening calibration got lucky). Each of the four is a unit test in the repository. Generation 0 is the operator’s config: the human’s settings win until the loop has promoted evidence-backed changes past them.

Empirical status. The architecture is shipped; its record is short. Over 18 daily meta-cycles (2026-07-10 to 09-06) the loop recorded 3 promotions, 1 rollback and 0 Brier vetoes (Table 7); of the candidates it scored, most (25 of 38) were refused for the thin-sample rule (held-in $n < 20$), which is the gate doing its job on a small window rather than the loop learning. The one promotion that reached the fresh window with a worse record than its parent was reverted two days later. We make no claim that the loop has improved trading: the active generation (min_edge 0.15, $w = 0.6$) scored $-\$3.00$ held-in against the champion’s $-\$6.22$, i.e. it lost less. Improvement axes are unified as mutation operators feeding the same accept mechanism: playbook rules and calibration refits, decision-parameter and prompt mutation (shipped), then workflow structure, then agent-plane code diffs (adding shadow trading and human diff review), then LoRA adapters with a time-discounted reward. The admission rule is strict: a parameter enters the space only when the gate can discriminate on it. *Tuning what you cannot measure is drift, not improvement.*

6 Leak-free evaluation methodology

The hardest part of evaluating a market agent is not computing PnL; it is not lying to yourself. hindcast teaches the baseline each station’s bias from archived as-of forecasts (the previous-runs API

of a public NWP aggregator) and official settlements, strictly *before* any window it is evaluated on. replay then backtests the *decision rule* against real historical order books, drawing prices from hourly candlesticks at a decision snapshot, and freezes every decision-time row (baseline probability, bid/ask at the snapshot, outcome) to a hashed file that every later experiment re-scores.

Snapshots are local, and the replay is asymmetric on purpose. The NWS CLI day is a *local* calendar day, so the snapshot is a local hour: for lows, local midnight, the first minute of the day, when nothing of the minimum can be on the books (a fixed 06Z, which we used first, was 2am on the east coast, two hours into the day); for highs, 11:00 local, when the afternoon peak is still a live contest. The governing rule, learned when an early backtest snapshotted the dawn low after it had been realized, is *never let the replayed market know something the replayed model does not*. The converse is allowed and is the case here: by 11:00 the book has already digested the morning’s observations, while the replayed model is stripped of intraday observations, same-day model runs, the LLM adjustment and the calibration layer. The replay therefore *handicaps* the model. Its numbers are a conservative bound on the decision rule, not a fair contest between the model and the crowd, and the “naked model” row of E1 must be read that way. Sensitivity to the snapshot hour is E3.

Official baselines. A learned station correction is a small MOS. The obvious question, “is it better than what the market already reads?”, is answered by pricing the same markets, with the same decision rule and the same out-of-window bias learning, from NBM and from GFS MOS bulletins archived as issued (the $d-1$ 12Z cycle, about a day ahead, matching the consensus’ lead-1 guidance) (E5).

Error bars. Markets on one day share a thermometer and a synoptic pattern, so trades are not independent. Every PnL-per-contract figure carries a 95% interval from a block bootstrap that resamples settlement *days*, a t -statistic on daily PnL, and the maximum drawdown of the daily path.

The gate uses the same pipeline. The row collector that scores a candidate generation is the one that enforces no-peeking, and it lives in the kernel where the optimizer cannot touch it.

7 Pre-registered experiments and results

Protocol. Window: settlement days from 2026-07-09 (the first day with archived lead-1 guidance for every station) to 2026-09-10 at the time of writing, rolling forward daily; ten U.S. stations, twelve Kalshi daily-temperature series (nine highs, three lows); station bias/ σ learned only from days before the window start; fees included (Kalshi taker: $\lceil 7\% \cdot p(1-p) \rceil$ cents per contract); production parameters min_edge = 0.08, $w = 0.5$, lead 1. Every table names the frozen dataset (file, SHA-256 prefix) it was scored on. Table 3 lists the experiments and, crucially, what each one *decides*; the decisions were written before the numbers.

Preliminary replay (v0, superseded). Before the protocol above was fixed, a 21-day replay (968 settled markets, 2026-06-24 to 07-14, UTC snapshots 06Z/15Z, no error bars) gave: naked model-vs-market $-\$0.007$ /contract over 425 trades (41% wins); +learned bias $-\$0.005$ over 396; +market-prior blend $+\$0.028$ over 184 (45%

#	Experiment	Decision rule (written before the data)
E1	Ablation: naked model $\rightarrow$ +station bias $\rightarrow$ +market blend $\rightarrow$ control : no blend, edge bar raised until it trades as rarely as the blend	If the control's PnL/ct is positive with a 95% CI excluding zero, <i>selectivity</i> is the edge. If only the blended row is, the edge is the <i>blend</i> (a better probability) and the thesis sentence changes to say so.
E2	Grid: min_edge $\times$ blend weight	The E1 conclusion must hold across neighbouring cells, not one.
E3	Snapshot sensitivity: lows at local 00/01/02, highs at 09/11	If PnL/ct moves by more than its CI half-width, timing is doing work and the leak-free claim is weakened accordingly.
E4	LLM on/off; clamp $\delta \in \{0, .05, .10, .15, .25, \infty\}$; PnL and Brier	Whether the bounded adjustment adds anything over the baseline, and at which δ ; Brier must not worsen.
E5	Same decision rule priced from NBM and GFS MOS, each with its own learned bias	If NBM ties the consensus, "learnable local edge" is dropped from the paper.
E6	Breakdown by station, ISO week, high/low	Whether the edge is broad or one station in one week.
E7	Sizing regimes on the production trades: flat, Kelly, Kelly+caps, Kelly+caps+kill-switch	What each rail costs or saves: the only numbers the "harness" layer gets.
E8	Self-improvement lineage over ≥ 4 weeks: promotions, rollbacks, vetoes	Whether Section 5 keeps an empirical claim or becomes future work.
E9	Station bias on first vs second half of the hindcast window	Whether "biases persist" is true (same sign, similar magnitude).

Table 3: Pre-registered experiments. Each is one command over the frozen dataset; results are regenerated daily and published with the data.

Rule	n	PnL/ct	95% CI	maxDD
Naked model vs market	1501	-0.041	[-0.058, -0.023]	61.8
+ learned station bias	1407	-0.037	[-0.056, -0.016]	55.5
+ market blend (prod.)	625	-0.034	[-0.069, +0.002]	23.7
Control: no blend, bar 0.19	618	-0.034	[-0.070, +0.002]	23.2

Table 4: E1 on the frozen dataset (window 2026-07-09 to 09-10, digest 7c3fb9f21ab727e3), 64 settlement days, 3,349 rows, win rates 37–38% throughout. CI: day-block bootstrap. maxDD: of the daily PnL path, \$ per one contract per trade.

wins). The blend more than halved activity and flipped expectancy positive. That is consistent with the thesis, but it is also consistent with the blend simply being a better probability, and 184 trades in one three-week window carries no error bar. It motivated E1's control and is not used for any conclusion below.

E1: ablation with the selectivity control. Table 4. Every row loses. The learned station bias helps a little ($-0.041 \rightarrow -0.037$); the blend helps a little more (-0.034) and halves activity; and the control, the un-blended model with its edge bar raised from 0.08 to 0.19 until it trades as rarely as the blend, lands on the *same* expectancy (-0.034 , $n = 618$ vs. 625) with an interval that includes zero on both. The pre-registered rule therefore returns neither answer: the blend and selectivity are one lever, and neither produces positive expectancy with this baseline. The v0 result (+\$0.028 over three weeks) does not reappear: the same weeks in this dataset are +0.004 (W28) and +0.074 (W30) against -0.066 (W29), and the 64-day mean is negative.

E2–E3: robustness. Of the 31 grid cells with at least 18 trades, all 31 are negative (range -0.029 to -0.140 /contract); the only positive cell ($w = 0.75$, min_edge 0.15) has six trades. The E1 conclusion is not a property of one cell. Snapshot sensitivity: moving the low

snapshot from local 00:00 to 01:00 or 02:00 changes production expectancy from -0.034 to -0.032 and -0.035 ; moving the high snapshot from 11:00 back to 09:00 changes it to -0.046 $[-0.083, -0.010]$. The largest shift (0.012) is a third of the E1 half-width (0.035), so timing is not doing the work. The direction of the 09:00 result matters more than its size: at 09:00 the book knows *less* about the day and the model loses *more*, so the model is not being beaten by a book that already saw the morning; it is beaten by a book that prices the same guidance better.

E4: does the LLM add anything? Table 5, on a 400-row deterministic stride sample of the frozen dataset (a reasoning model costs about a minute per row). The bounded adjustment does not worsen calibration: Brier improves monotonically from the baseline's 0.1706 to 0.1691 at $\delta = 0.15$ and 0.25, and the *unclamped* model ($\delta = \infty$) gives back part of that (0.1694). The production clamp is at the best of the sweep, which is the harness thesis in miniature: the model helps a little when it may only nudge, and helps less when it may replace. The effect is small (-0.0015 Brier, about 1%) and the PnL movement ($+0.009 \rightarrow +0.020$ /contract on ~ 78 trades) is well inside a ± 0.10 interval; on this sample the baseline itself is slightly positive, which is the sampling noise the full-dataset E1 warns about. The live paper run tells a different story (Section 7): the live forecaster's Brier over its 41 *traded* markets is 0.43 against a market Brier of 0.10–0.19 at the same stations. The two numbers are not the same quantity (replay scores every row; live scores only the markets the agent chose to trade, a selection biased toward its largest disagreements with the market), but the gap says where the damage is done: not in the clamped adjustment, but in the trade selection downstream of it.

E5: against the official guidance. On the same 3,349 markets, priced by the same rule with each source's own out-of-window bias: the seven-model consensus scores Brier 0.1627, NBM 0.1631, GFS MOS 0.1651 ($n = 3,344$ paired); PnL/contract -0.034 , -0.055 $[-0.088, -0.021]$, and -0.035 $[-0.069, +0.001]$. The consensus with learned bias ties NBM. **We drop the "learnable local edge" claim:**

Clamp δ	n	PnL/ct	95% CI	Brier
0 (baseline, no LLM)	79	+0.009	[−0.087, +0.098]	0.1706
± 0.05	77	+0.016	[−0.089, +0.121]	0.1698
± 0.10	77	+0.016	[−0.089, +0.121]	0.1694
± 0.15 (production)	78	+0.020	[−0.082, +0.123]	0.1691
± 0.25	79	+0.019	[−0.081, +0.121]	0.1691
∞ (unclamped)	79	+0.019	[−0.081, +0.121]	0.1694

Table 5: E4: LLM-in-replay on 400 stride-sampled rows of the frozen dataset (digest 7c3fb9f21ab727e3): a self-hosted DeepSeek reasoning model, temperature 0, one sample per row, prompt rebuilt from as-of data only; its probability is clipped to baseline $\pm \delta$ and pushed through the production decision rule. Brier is over all 400 rows; PnL over the trades the rule takes.

Sizing	Return	maxDD	Entered	Halted
Flat 1%	−10.2%	22.8%	88	—
Kelly 0.15×	−23.3%	70.5%	88	—
Kelly + caps	−41.5%	77.2%	88	—
Kelly + caps + kill-switch	+12.0%	15.2%	11	07-17

Table 6: E7: the same 625 trades under each sizing regime (conservative profile: Kelly 0.15×, caps 5/10/25/50%, kill-switch at 15% drawdown; 537 of the trades fail the live engine’s fee-aware edge check at their snapshot price and are never entered). “Entered” counts positions opened.

the correction is real (E9) but it recovers what the NWS’s own blend already publishes, and the market reads that.

E6: where the edge lives. Four of ten stations are positive (Chicago +0.071, $n = 59$; Miami +0.040; New York +0.033; San Antonio +0.015, $n = 29$) and six negative (Atlanta −0.147, Denver −0.093, Austin −0.065, Los Angeles −0.064, Philadelphia −0.049, Dallas −0.022); three of ten weeks are positive. Lows (-0.006 , $n = 71$) lose less than highs (-0.038 , $n = 554$), consistent with the local-midnight snapshot leaving the low a genuinely open contest while the 11:00 book already knows the morning. None of the positive stations’ intervals exclude zero on their own.

E7: the harness, scored. The 625 production trades through the risk engine on a \$1,000 paper bankroll (Table 6). With a losing rule, sizing is a multiplier on the loss: flat 1% stakes lose 10%, fractional Kelly 23% with a 71% drawdown, and the concentration caps change which losses are taken rather than whether (−42%: they cut the winners’ size on multi-strike days as much as the losers’). The kill-switch is the rail that mattered: it halted trading on 2026-07-17 after 11 entries, refused the next 547, and ended the window at +12% with a 15% drawdown, which is the honest reading of “the harness is the edge” on this data: the harness did not create an edge, it stopped a losing rule from compounding.

E8–E9. The lineage is Table 7. Station bias persists: on lead-1 highs and lows, 11 of 12 (station, kind) biases keep their sign between the first 123 verified days and the last 32 (the exception,

gen	par.	by	edge	w	in	out	status
0	—	oper.	0.08	0.5	—	—	baseline
1	0	LLM	0.12	0.6	+2.31	+1.63	prom. 07-23
2	1	rand.	0.115	0.59	+9.22	+1.16	07-25, reverted
3	1	LLM	0.15	0.6	−3.00	−1.32	09-01, active

Table 7: E8: self-improvement lineage as exported by the loop (18 meta-cycles, 2026-07-10 to 09-06). edge: min_edge; w: market-prior weight; in/out: held-in and held-out total PnL (\$) at promotion. Generation 2 was reverted when its parent beat it on the rolled-forward window.

Chicago Midway, is −0.1 vs. +0.3, i.e. zero both times), and magnitudes hold (Miami +2.0 $\rightarrow$ +1.8, Philadelphia +2.6 $\rightarrow$ +1.6, Atlanta +2.4 $\rightarrow$ +2.1, LAX −1.0 $\rightarrow$ −1.2, Austin low −2.1 $\rightarrow$ −3.8). The correction is real; E5 says the market already has it.

The live paper run. The reference agent has paper-traded continuously since July (1,911 cycles, \$100,000 allocated) with every live-only advantage on: intraday observations, same-day model runs, the LLM adjustment, calibration. At the time of writing: 41 settled trades, realized −\$1,679, unrealized +\$593, total −1.1%, maximum drawdown 3.0%, win rate 24% with average win \$243 against average loss \$132. The asymmetry is the harness’s; the expectancy is not there. Its forecaster’s Brier (0.43) against the market’s (0.10–0.19) is the sharpest single number in this paper.

8 Discussion and limitations

The claims are only as wide as the window. The replay is paper mode at real bid/ask; it is an ablation over one summer season, not a track record. The live paper run is the ongoing public test, reported with wins and losses alike, and this paper’s tables are regenerated from it daily.

What the replay cannot see. Replay has no news brief, no intraday observations, no forecaster-discussion text, and (except in E4) no LLM and no calibration layer; it measures the decision rule over the statistical baseline, the dominant live pathway for weather markets, but not the whole live agent. A residual leak risk, an LLM “remembering” recent weather from pre-training, is negligible with a local model whose cutoff precedes the window, and must be rechecked whenever the forecaster is swapped.

Capacity is the real ceiling. Daily-temperature markets are thin. A survey of seven Kalshi cities found on the order of a few hundred thousand dollars of daily notional in aggregate, which bounds extraction at low single-digit millions per year even under optimistic assumptions. This is a methodology contribution about *measurable* edge, not a claim of scalable returns.

Generalization. The strong-evaluator property is what makes weather the right first domain; it is also what most other markets lack. Extending the harness and the gate to weaker-evaluator markets is future work; the journal already archives each live forecast’s inputs so a trace-replay evaluator can begin to close that gap.

9 Reproducibility, ethics, and broader impact

The system, the evaluation pipeline, the frozen replay datasets with their digests, the experiment results, and the as-of forecast record (NWP guidance, settlements, own GraphCast/GenCast rows, NBM/MOS baselines) are public and updated daily; paper mode requires no exchange keys and reproduces every table from the published files (Appendix A). Live trading is off by default and requires two independent explicit opt-ins; the agent has no deposit, withdrawal, or transfer capability. We ship hard position limits and a drawdown kill-switch, and we state plainly that no software makes trading safe: roughly 70% of Polymarket addresses have lost money, and every layer of this paper is about losing less. On the self-improvement side, the safety argument is architectural rather than behavioral: the judge is off the optimizer’s edit surface by construction, safety rails are outside the parameter space by a kernel policy test, and every promotion is an auditable commit authored by the loop itself.

10 Conclusion

Autonomous LLM traders have so far lost money, and the profile of the runs that lost least points at discipline rather than forecasting. OpenThomas puts the discipline in deterministic code, uses a language model only for a bounded adjustment, and points the whole system at weather markets, the one category where the resulting edge can be measured densely, objectively, and without leakage. The contribution we are surest of is the measurement: explicit windows, frozen and hashed datasets, local-time snapshots, official baselines, day-block error bars, and a pre-registered control that can falsify our own thesis. On 64 days of data the pre-registered test came back against us: no setting of the baseline-only rule has positive expectancy, the blend and the edge bar are the same lever, and the learned station correction is what the NWS blend already publishes. What the harness demonstrably did was limit the damage: the kill-switch preserved capital, the gate refused thin evidence and reverted an overfit promotion. A positive edge in this market, if it exists, needs what the replay strips out and the live agent has not yet turned into calibration: intraday observations, same-day runs, and a language-model adjustment that keeps doing on live inputs what it does on replay rows: E4 shows a clamped model improving calibration slightly and an unclamped one giving part of it back, while the live forecaster’s Brier says the loss is made downstream, in which markets get traded. We report the result in public, losses included, and invite the community to check it.

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A Reproducibility details

Item	Value
Stations	KNYC, KAUS, KMDW, KMIA, KLAX, KPHL, KDEN, KDFW, KATL, KSAT
Series	9 daily-high, 3 daily-low Kalshi series
NWP guidance	Open-Meteo previous-runs API, hourly 2 m temperature, models gfs, ecmwf_ifs025, icon, gem, ukmo, meteofrance, jma; daily extremes over the local day; leads 1–5
Settlements	ACIS daily max/min for the NWS station (the CLI integer)
Official baselines	IEM MOS archive, models NBS (txn) and GFS MOS (n_x), $d-1$ 12Z cycle
Bias/ σ	shrinkage pseudo-count 10 toward $\sigma_{\text{prior}}(\ell) = \{2.3, 2.8, 3.4, 4.0, 4.6\}^\circ\text{F}$ for $\ell = 1..5$; learned only from days < window start
Strike probability	$0.8 \mathcal{N}(\hat{T}, \sigma) + 0.2 \mathcal{N}(\hat{T}, 2\sigma)$ at half-integer thresholds
Snapshot	lows: local 00:00; highs: local 11:00; hourly candle close bid/ask; spread ≤ 0.10 required
Decision rule	$p = w \text{mid} + (1-w)\hat{p}$, $w = 0.5$; trade the side with $p - \text{price} - \text{fee} \geq 0.08$, price in $[0.05, 0.95]$, at most one side per market
Fees	Kalshi taker $\lceil 7p(1-p) \rceil$ cents per contract
Error bars	day-block bootstrap, 2,000 resamples, seed 0
Risk profile (E7)	conservative: Kelly 0.15 $\times$, caps 5/10/25/50%, kill-switch 15%; \$1,000 bankroll
LLM (E4)	DeepSeek (self-hosted, OpenAI-compatible endpoint); temperature 0, single sample, prompt cached by exact text; 400-row deterministic stride sample
Commands	hindcast -days 92; replay -start -end -rows-out; ablate -rows-in -out

Table 8: Everything needed to regenerate the tables from the published files.